MATTHEW H. ZHANG

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EDUCATION

The University of Texas at Austin – GPA: 3.73 **May 2028**

Bachelor of Science in Electrical and Computer Engineering Honors

Bachelor of Business Administration in Canfield Business Honors

Relevant Coursework: Circuit Analysis & Design Honors, Computing Systems Honors, Embedded Systems Honors, Software Design & Implementation Honors, Data Science Honors, Management Information Systems Honors

EXPERIENCE

***Hardware & Venture Analyst Intern, Humanoid Teleoperation Collaboration* – Tess Ventures** **Mar 2026 - Present**

- Led development of a wearable rig capturing human arm joint angles at 50 Hz via custom binary packet
- Transmitted motion data over **USB serial** to teleoperate a **Unitree G1 EDU** humanoid
- Designed modular MCU-based magnetic encoder PCB using **RS-485 UART** for scalable joint-angle sensing
- Prepared **technical demos and product-facing presentations** for engineering and business audiences

***Control Systems & Electrical Engineer* – Texas Aerial Robotics** **Jan 2026 - Present**

- Modeled inverted pendulum and TVC system dynamics and implemented **PID, MPC, & LQR** controllers in Python
- Designing a compact AM32-based **30A 4S BLDC motor driver PCB** under size and thermal constraints in Altium

***Embedded Systems Engineer* – GHOST Robotics** **Nov 2025 - Present**

- Validated custom sensor PCBs and integrated them into tight mechanical assemblies using Onshape
- Redesigned I2C sensor address-splitter PCB, **reducing board area by 40%**, improving connector placement
- Performed board bring-up and debug by validating power rails and signals with a multimeter and logic analyzer
- Wrote interrupt-based **RP2040 sensor polling firmware** in C with custom sensor drivers over USB serial

***Team Captain, Lead Engineer* – BagelBytes Robotics** **Dec 2024 - Aug 2025**

- Organized a team of **30+** individuals to successfully design, manufacture, and field functioning 125 lb robots
- Mentored **20+** members in CAD & electrical systems; **improved team ranking by 1,500+ places** during leadership
- Engineered custom **CAN bus** layout and architecture, reducing match-time electrical failures by **28%**
- Designed **100+ custom sheet metal parts** and 7 custom-built gearboxes with Onshape

PROJECTS

Personal Project Vinyl Record Player **Dec 2025 - Present**

- Independently designed MCU-based motor driver 2-layer PCB with **USB-C PD** power negotiation for turntable
- Created power distribution with custom **buck converter** (12V → 5V), local decoupling, differential-pair routing

1st Place, $6000 Prize - UT Genesis x Momentum x Yconic Hackathon **April 2026**

- Built an ESP32 **wearable AI assistant** with face recognition, live STT, and persistent memory of conversations
- Developed a pipeline for face tracking, identity matching, and context retrieval across repeated interactions

1st Place - IEEE Robotics & Automation Society, Robotathon Competition **Sep 2025 - Nov 2025**

- Led electromechanical design of autonomous competition robot that launched balls with custom flywheels
- Adjusted designs for modularity, allowing for easy sensor iteration and improved maze routing

SKILLS

Technical Skills: Circuit Design, PCB Layout, Firmware Development, CAD, Git, Debugging, Control Systems, Soldering

Tools: KiCad, Altium, LTSpice, MuJoCo, Linux, Onshape, GDB, Oscilloscope, Logic Analyzer, Excel, Machining & Shop Tools

Languages: C, C++, Python, Assembly, R, SQL | **Embedded:** STM32Cube, ESP-IDF, CCSTUDIO, I2C, SPI, UART, BLE

Yconic Hackathon 1st Place - $6000 (2026), IEEE RAS Robotathon 1st Place, Software Design Award (2025), FIRST Robotics Rising All Stars Award (2025), World's Rookie Inspiration Award, FIRST Robotics Worlds Qualifier (2024)